

Repeating Decimals to FractionsAdditional Practice 7A.01.02 **ANSWER KEY****Use the repeating decimal 0.5888... for questions 1 through 4.**

1. Let $x = 0.5888\ldots$. Complete each equation.

$$10x = 5.\overline{8}$$

$$100x = 58.\overline{8}$$

Only one digit repeats, and there is a 5 in front of it that does not. $10 \times 0.5888\ldots = 5.888\ldots$ and $100 \times 0.5888\ldots = 58.888\ldots$

2. Which two values, when subtracted, leave a whole number? Fill in one bubble.

- ☐ 10x and x
☐ 100x and x
☒ **100x and 10x**

$58.888\ldots - 5.888\ldots$ leaves a whole number. Using x instead of $10x$ would leave $58.888\ldots - 0.5888\ldots$, which still has a decimal tail.

3. Use two of the equations from question 1 to find the fraction equivalent of 0.5888... Show the subtraction step.

$$100x - 10x = 58.888\ldots - 5.888\ldots$$

$$90x = 53$$

$$x = \frac{53}{90}$$

53 is prime, so 53/90 will not simplify.

4. Write $3.\overline{58}$ as a fraction and as a mixed number.

Fraction: $\frac{323}{90}$

Mixed number: $3\frac{53}{90}$

$$3 \times 90 + 53 = 323.$$

A trail mix recipe uses these amounts: raisins $21.\overline{6}\%$, peanuts $\frac{5}{9}$, chocolate chips 0.14, and the rest is chex mix.

The bar means the 6 repeats forever.

5. What fraction of the recipe is chocolate chips?

$$\frac{7}{50}$$

$0.14 = 14/100$, which simplifies to $7/50$.

6. What fraction of the recipe is raisins?

$$\frac{13}{60}$$

$21.\overline{6}\% = 0.21666\dots$ TWO digits sit in front of the repeat, so move past both: $1000x - 100x$ gives $900x = 195$, and $x = 195/900 = 13/60$. ($100x - 10x$ would give $90x = 19.5$, and 19.5 is not a whole number.)

7. What is the decimal equivalent of the peanuts?

$$0.\overline{5}$$

$5 \div 9 = 0.5555\dots$ Use the bar.

8. About what percent of the recipe is chex mix? Round to the nearest whole percent.

9%

Over 900ths: $13/60 = 195/900$, $5/9 = 500/900$ and $7/50 = 126/900$, which total $821/900$. Then $900 - 821 = 79$, and $79 \div 900 = 0.0877\dots$, which rounds to 9%.

For questions 9 and 10, complete the table.

	Decimal	Percent	Fraction or Mixed Number
9.	$9.1\overline{52}$	$915.\overline{25}\%$	$9\frac{151}{990}$
10.	$0.01\overline{6}$	$1.\overline{6}\%$	$\frac{1}{60}$

9) For the fraction, one digit does not repeat, so use $1000x - 10x$: that gives $990x = 9061$, and $9061/990 = 9\frac{151}{990}$. 10) Divide by 100 first: $1.666\dots\% = 0.01666\dots$ Two digits sit in front of the repeat, so $1000x - 100x$ gives $900x = 15$, and $x = 15/900 = 1/60$.